CHAPTER 12

Ecosystem

Ecosystem-Structure And Function

1. Which ecosystem has the maximum biomass? *(2017-Delhi)*

a. Forest ecosystem
b. Grassland ecosystem
c. Pond ecosystem
d. Lake ecosystem

2. The term ecosystem was coined by: *(2016 - I)*

a. E.P. Odum
b. A.G. Tansley
c. E. Haeckel
d. E. Warming

3. Which one of the following is a characteristic feature of cropland ecosystem? *(2016 - I)*

a. Absence of soil organisms
b. Least genetic diversity
c. Absence of weeds
d. Ecological succession

4. Vertical distribution of different species occupying different levels in a biotic community is known as: *(2015)*

a. Zonation
b. Pyramid
c. Divergence
d. Stratification

5. An association of individuals of different species living in the same habitual and having functional interactions is: *(2015 Re)*

a. Biotic community
b. Ecosystem
c. Population
d. Ecological niche

Productivity

6. In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is $100x$ (kcalm^{-2}) yr^{-1}, what would be the GPP (Gross Primary Productivity') of the third trophic level of the same ecosystem? *(2024)*

a. $10x\ (kcalm^{-2})\ yr^{-1}$
b. $\frac{100x}{3x}\ (kcalm^{-2})\ yr^{-1}$
c. $\frac{x}{10}(kcalm^{-2})\ yr^{-1}$
d. $x\ (kcalm^{-2})\ yr^{-1}$

7. In the equation GPP – R = NPP *(2023)*
GPP is Gross Primary Productivity
NPP is Net Primary Productivity.
R here is

a. Reproductive allocation
b. Photosynthetically active radiation
c. Respiratory quotient
d. Respiratory loss

8. The amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis is called: *(2022 Re)*

a. Net primary production
b. Secondary production
c. Primary production
d. Gross primary production

9. In the equation GPP- R = NPP. *(2021)*
R represents:

a. Retardation factor
b. Environment factor
c. Respiration losses
d. Radiant energy

10. In relation to gross primary productivity and net primary productivity of an ecosystem, which one of the following statements is correct? *(2020)*

a. Gross primary productivity is always more than net primary productivity.
b. Gross primary productivity and net primary productivity are one and same.
c. There is no relationship between Gross primary productivity and net primary productivity
d. Gross primary productivity is always less than net primary productivity.

11. In an ecosystem, the rate of production of organic matter during photosynthesis is termed as: *(2015)*

a. Secondary productivity
b. Net productivity
c. Net primary productivity
d. Gross primary productivity

12. Secondary productivity is rate of formation of new organic matter by: *(2013)*

a. Decomposer
b. Producer
c. Parasite
d. Consumer

Decomposition

13. Identify the correct statements: *(2023)*

A. Detrivores perform fragmentation
B. The humus is further degraded by some microbes during mineralization.
C. Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.
D. The detritus food chain begins with living organisms.
E. Earthworms break down detritus into smaller particles by a process called catabolism.

Choose the correct answer from the options given below:

a. D, E, A only
b. A, B, C only
c. B, C, D only
d. C, D, E only

14. Given below are two statements: *(2022)*

Statement I: Decomposition is a process in which the detritus is degraded into simpler substances by microbes.

Statement II: Decomposition is faster if the detritus is rich in lignin and chitin

In the light of the above statements, choose the correct answer from the options given below.

a. Statement I is incorrect but Statement II is correct
b. Both Statement I and Statement II are correct
c. Both statement I and statement II are incorrect
d. Statement I is correct but Statement II is incorrect

15. Detritivores breakdown detritus into smaller particles. This process is called. *(2022)*

a. Decomposition b. Catabolism
c. Fragmentation d. Humification

16. The rate of decomposition is faster in the ecosystem due to following factors EXCEPT: *(2020-Covid)*

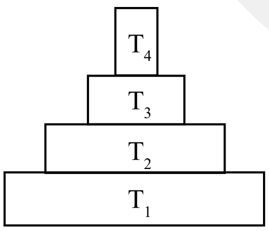
a. Warm and moist environment
b. Presence of aerobic soil microbes
c. Detritus richer in lignin and chitin
d. Detritus rich in sugars

17. Which one of the following processes during decomposition is correctly described? *(2013)*

a. Leaching: Water soluble inorganic nutrients rise to the top layers of soil
b. Fragmentation: Carried out by organisms such as earthworm
c. Humification: Leads to the accumulation of a dark colored substance humus which undergoes microbial action at a very fast rate
d. Catabolism: Last step in the decomposition under fully anaerobic condition

Energy Flow & Ecological Pyramids

18. Consider the pyramid of energy of an ecosystem given below: *(2024 Re)*



If T_4 is equivalent to 1000 J, what is the value at T_1?

a. $\frac{10000}{10}$ J b. $\frac{10000}{10} \times 4$J c. 10,000 J d. 10,00,000 J

19. Which one of the following is **not** a limitation of ecological pyramids? *(2024 Re)*

a. Saprophytes are not given any place in ecological pyramids.
b. It assumes a simple food chain, that almost never exists in nature.
c. It accommodates a food web.
d. It does not take into account the same species belonging to two or more trophic levels.

20. Which of the following statements is not correct ? *(2021)*

a. Pyramid of biomass in sea is generally upright.
b. Pyramid of energy is always upright.
c. Pyramid of numbers in a grassland ecosystem is upright.
d. Pyramid of biomass in sea is generally inverted.

21. Match the trophic levels with their correct species examples in grassland ecosystem. *(2020)*

	Column-I		Column-II
1.	Fourth trophic level	(i)	Crow
2.	Second trophic level	(ii)	Vulture
3.	First trophic level	(iii)	Rabbit
4.	Third trophic level	(iv)	Grass

a. 1-(iii) 2-(ii) 3-(i) 4-(iv)
b. 1-(iv) 2-(iii) 3-(ii) 4-(i)
c. 1-(i) 2-(ii) 3-(iii) 4-(iv)
d. 1-(ii) 2-(iii) 3-(iv) 4-(i)

22. Which of the following statements is incorrect? *(2020-Covid)*

a. Energy content gradually increases from first to fourth trophic level
b. Number of individuals decreases from first trophic level to fourth trophic level
c. Energy content gradually decreases from first to fourth trophic level
d. Biomass decreases from first to fourth trophic level

23. Which of the following ecological pyramids is generally inverted? *(2019)*

a. Pyramid of numbers in grassland
b. Pyramid of energy
c. Pyramid of biomass in a forest
d. Pyramid of biomass in a sea

24. What type of ecological pyramid would obtained with the following data? *(2018)*

Secondary consumer : 120 g
Primary consumer : 60 g
Primary producer : 10 g

a. Inverted pyramid of biomass
b. Pyramid of energy
c. Upright pyramid of numbers
d. Upright pyramid of biomass

25. The primary producers of the deep-sea hydrothermal vent ecosystem are: *(2016 - II)*

a. Blue-green algae b. Coral reefs
c. Green algae d. Chemosynthetic bacteria

26. The mass of living material at a tropic level at a particular time is called: *(2015)*

a. Net primary productivity
b. Standing crop
c. Gross primary productivity
d. Standing state

27. Most animals that live in deep oceanic waters are: *(2015)*

a. Secondary consumers b. Tertiary consumers
c. Detritivores d. Primary consumers

28. If 20 J of energy is trapped at producer level, then how much energy will be available to peacock as food in the following chain? *(2014)*

Plant → Mice → Snake → Peacock

a. 0.0002 J b. 0.02 J c. 0.002 J d. 0.2 J

Answer Key

1. (a)	**2.** (b)	**3.** (b)	**4.** (d)	**5.** (a)	**6.** (a)	**7.** (d)	**8.** (c)	**9.** (c)	**10.** (a)
11. (d)	**12.** (d)	**13.** (b)	**14.** (d)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (c)	**20.** (a)
21. (d)	**22.** (a)	**23.** (d)	**24.** (a)	**25.** (d)	**26.** (b)	**27.** (c)	**28.** (b)		